

| Threat Category               | Attack Description                                                                             | Mitigation                                                                                                                             |
|-------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Spoofing</b>               | An attacker tries to impersonate ESP32 to inject fake power readings into the database.        | The Mobius server requires a strict SOrigin identity header. Mosquitto also requires valid SSL certificates to establish a connection. |
| <b>Tampering</b>              | A malicious user intercepts the WiFi signal to alter the voltage numbers in transit.           | All data is wrapped in TLS 1.2 encryption. Any intercepted packets are mathematically scrambled and cannot be modified.                |
| <b>Repudiation</b>            | The system fails to prove exactly when a specific data packet was created or received.         | Mobius automatically generates and logs a strict creation timestamp for every single Content Instance inside the container.            |
| <b>Information Disclosure</b> | An unauthorized user snoops the local network to read private energy usage habits.             | Verified via Wireshark capture. The Application Data is completely unreadable ciphertext to anyone sniffing the network.               |
| <b>Denial of Service</b>      | An external attacker floods the server with fake traffic to crash the energy monitor.          | The system is isolated on a private mobile hotspot routing environment. It is not exposed to the public internet.                      |
| <b>Elevation of Privilege</b> | A standard user tries to delete the database or alter the Application Entity folder structure. | Strict oneM2M access control policies ensure only the authorized administrator can build or modify the core server containers.         |